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Practice session 3: decision trees

Q1 Consider a dataset for predicting the outcome of a weekend event, which has the following class distribution of activities:

Class	# Instances
Cinema	15
Tennis	5
Shopping	5
Stay-in	5

Calculate the Gini index for this dataset.

Provide your answer rounded to three decimal places. 0.667

$GI = 1 - \sum p_i^2$

$= 1 - \left(\left(\frac{15}{30}\right)^2 + \left(\frac{5}{30}\right)^2 + \left(\frac{5}{30}\right)^2 + \left(\frac{5}{30}\right)^2\right)$

$= 1 - \left(\frac{1}{4} + \frac{1}{36} \cdot 3\right) = 1 - \left(\frac{1}{4} + \frac{1}{12}\right) = 1 - \frac{16}{48} = 1 - \frac{1}{3} = \frac{2}{3}$

$= 0.667$

Q2 Consider a classification task with five distinct classes. Calculate the maximum Gini index for this dataset. 0.8

max. if class distribⁿ is uniform.

$GI = 1 - \left(\left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^2\right)$ or $\frac{K-1}{K}$

$= 1 - \frac{1}{5} = \frac{4}{5}$

Q3 Consider a dataset containing 100 data points. A bootstrap sample is taken by randomly selecting 100 data points with replacement. Calculate the expected Out of Bag (OOB) fraction, i.e., the expected proportion of the original dataset that is not included in the bootstrap sample.

A) 0.37 ✓
B) 0.63
C) 1.0
D) 0.50

$x_1, x_2, \ldots, x_{100}$

P(Data point is not chosen in single draw) $= 1 - \frac{1}{100} = 0.99$

P(Data point will not be chosen in 100 draws) $= (0.99)^{100} \approx 0.366$

Q4 Consider an AdaBoost scenario where you have a dataset consisting of four data points, each with an initial weight of $\frac{1}{4}$. A weak learner, h_1, classifies these points, resulting in the following predictions:

- $h_1(x_1) = y_1$ (correct)
- $h_1(x_2) \neq y_2$ (incorrect)
- $h_1(x_3) = y_3$ (correct)
- $h_1(x_4) \neq y_4$ (incorrect)

Given that the total error ϵ_1 of this learner is calculated to be 0.5, calculate the performance weight α_1 for this learner. Provide your answer rounded to three decimal places. 0

error $= \epsilon_1 = 0.5$

$\Rightarrow \alpha_t = \frac{1}{2} \ln\left(\frac{1-\epsilon_t}{\epsilon_t}\right) = \frac{1}{2} \ln\left(\frac{0.5}{0.5}\right) = 0$

$\alpha_1 = 0$

Q5 Which of the following statements is true regarding the use of Gini Index and Entropy as splitting criteria in the construction of decision trees?

A) The Gini Index is computationally more intensive than Entropy because it requires logarithmic calculations.
B) Both Gini Index and Entropy will always result in the same tree structure as long as the dataset does not change.
C) Gini Index tends to favor larger partitions while Entropy is more sensitive to changes in class probabilities within nodes. ✓
D) Entropy can result in more balanced trees compared to the Gini Index because it maximizes information gain.

$GI = 1 - \sum p_i^2$
Entropy $= -\sum p_i \log p_i$

↳ Both max. IG.

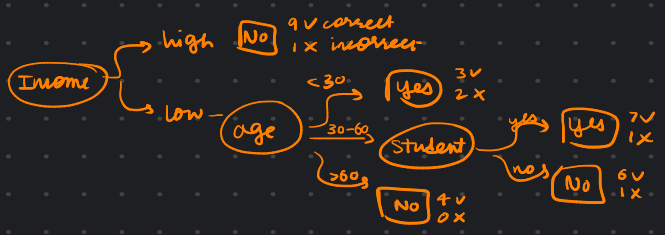
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Imagine a decision tree used to predict whether a person buys a certain product or not, based on three features: Age, Income, and Student Status. The tree is structured as follows:

1. **Root Node (Income)**
   - High Income
     - Predict: No (Leaf Node, 10 instances, 1 misclassified)
   - Low Income
     - **Node (Age)**
       - Below 30
         - Predict: Yes (Leaf Node, 5 instances, 2 misclassified)
       - 30 to 60
         - **Node (Student Status)**
           - Student
             - Predict: Yes (Leaf Node, 8 instances, 1 misclassified)
           - Not a Student
             - Predict: No (Leaf Node, 7 instances, 1 misclassified)
       - Above 60
         - Predict: No (Leaf Node, 4 instances, 0 misclassified)

The total number of instances in the training data is 34. The initial misclassification rate for the entire tree is calculated based on misclassifications at the leaf nodes.

Question: Calculate the initial misclassification rate for the entire tree.



Total instances classified = 34

of misclassifications = 5

misclassification rate = $\frac{5}{34} = 0.15$

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Question: Assuming that pruning the "Age" node for "Below 30" reduces the number of leaf nodes and total misclassifications to 4 and ~~4~~ 6, respectively. If the cost complexity pruning parameter that would justify this pruning is α. Calculate α x 10^4 rounded to nearest integer

	initial	final
# of leaves	5	4
# of misclassifications	5	~~4~~ 6

$R_\alpha(T) = R(T) + \alpha|T|$

Initial misclassification rate = $\frac{5}{34} + 5\alpha$

Updated misclassification rate = $\frac{6}{34} + 4\alpha$

To justify, Updated $R_\alpha(T)$ < Initial $R_\alpha(T)$

$\frac{6}{34} + 4\alpha < \frac{5}{34} + 5\alpha$

$\alpha > +\frac{1}{34}$

$\alpha_{min} = 0.02941$

$= 294.1 \times 10^4$

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In the context of ensemble learning, which of the following statements correctly describe the characteristics and benefits of using Random Forests and Boosting techniques?

A) Random Forests reduce variance in the model predictions by averaging multiple deep decision trees trained on different subsets of the data. ✓
B) Boosting can increase the risk of overfitting as it focuses on increasing the accuracy of the ensemble by sequentially correcting the errors of weak learners. ✓
C) In Boosting, each model is built independently, whereas, in Random Forests, models are built sequentially to correct the errors of the previous model. ✗
D) Both Random Forests and Boosting typically yield better results when combined with low-bias, high-variance classifiers. ✓

Q9

You have a friend who only does one of four things on every Saturday afternoon: go shopping, watch a movie, play tennis, or just stay in. Over 11 different weekends, you've collected data on various factors that could influence your friend's choice of activity. These factors include the weather conditions (sunny, windy, or rainy), whether the parents visit, the financial condition after withdrawing cash (rich or poor), and whether an exam is scheduled for the upcoming week. Based on this data, you aim to build a decision tree to predict the activity of your friend on any future Saturday afternoon.

The data collected is summarized in the table below:

#	Weather	Parents	Cash	Exam	Decision (y)
1	Sunny	Visit	Rich	Yes	Cinema
2	Sunny	No-visit	Rich	No	Tennis
3	Windy	Visit	Rich	No	Cinema
4	Rainy	Visit	Poor	Yes	Cinema
5	Rainy	No-visit	Rich	No	Stay-in
6	Rainy	Visit	Poor	No	Cinema
7	Windy	No-visit	Poor	Yes	Cinema
8	Windy	No-visit	Rich	Yes	Shopping
9	Windy	Visit	Rich	No	Cinema
10	Sunny	No-visit	Rich	No	Tennis
11	Sunny	No-visit	Poor	Yes	Tennis

Question: Calculate the Gini index for the initial state of the decision-making dataset regarding the weekend activities of your friend

(Provide your answer rounded to two decimal places.)

cinema → 6
tennis → 3
stay-in → 1
shopping → 1

$$GI = 1 - \left(\left(\frac{6}{11}\right)^2 + \left(\frac{3}{11}\right)^2 + \left(\frac{1}{11}\right)^2 + \left(\frac{1}{11}\right)^2\right) = 1 - \frac{97}{121} = \frac{24}{121} = 0.6115 \approx 0.61$$

Q10

Question: What is the entropy of the initial state of the decision-making dataset regarding the weekend activities of your friend?

(Provide your answer rounded to two decimal places.)

$$H(D) = -\sum p_i \log p_i = -\left(\frac{6}{11}\log\frac{6}{11} + \frac{3}{11}\log\frac{3}{11} + \frac{1}{11}\log\frac{1}{11} + \frac{1}{11}\log\frac{1}{11}\right) \approx 1.617 \approx 1.62$$

Q11

Question: Calculate the information gain for splitting the data based on the "Weather" attribute.

(Provide your answer rounded to two decimal places.)

Initial entropy = 1.617

After splitting on weather:

sunny	windy	rainy
cinema → 1	cinema → 3	cinema → 2
tennis → 3	shopping → 1	stay-in → 1

Entropy:

$$-\left(\frac{1}{4}\log\frac{1}{4} + \frac{3}{4}\log\frac{3}{4}\right) = 0.811$$

$$\approx 0.811$$

$$-\left(\frac{2}{3}\log\frac{2}{3} + \frac{1}{3}\log\frac{1}{3}\right) \approx 0.918$$

weighted entropy $= \frac{4}{11}(0.811) + \frac{4}{11}(0.811) + \frac{3}{11}(0.918) \approx 0.8402$

IG = Initial entropy − Final entropy

IG = 1.617 − 0.8402 = 0.7768 ≈ 0.78

Q12

Question: Based on the weekend activities data of your friend and the factors affecting the decision (Weather, Parents, Cash, and Exam), identify which of the following statements are true regarding the decision-making patterns:

A) When it is rainy and there is no exam, the decision is always to watch a movie. ✗
B) Whenever the weather is windy, no visit from parents leads to a diverse set of activities, making it a less predictive factor. ✓
C) The financial condition ("Cash" attribute) significantly affects the decision when there is no parental visit and the weather is sunny. ✗
D) An exam during the upcoming week combined with a visit from the parents often results in choosing to watch a movie, regardless of the weather condition. ✓

#	Weather	Parents	Cash	Exam	Decision (y)
1	Sunny	Visit	Rich	Yes	Cinema
2	Sunny	No-visit	Rich	No	Tennis
3	Windy	Visit	Rich	No	Cinema
4	Rainy	Visit	Poor	Yes	Cinema
5	Rainy	No-visit	Rich	No	Stay-in
6	Rainy	Visit	Poor	No	Cinema
7	Windy	No-visit	Poor	Yes	Cinema
8	Windy	No-visit	Rich	Yes	Shopping
9	Windy	Visit	Rich	No	Cinema
10	Sunny	No-visit	Rich	No	Tennis
11	Sunny	No-visit	Poor	Yes	Tennis

→ Tennis regardless of rich/poor when sunny, no-visit

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Consider a dataset used for a classification task with several different scenarios regarding the distribution of instances across various classes. Identify which of the following statements are correct about the entropy in these scenarios:

A) A dataset with four classes, each containing an equal number of instances, will have an entropy of 2. ✓
B) If all instances in a dataset fall into a single class, the entropy of the dataset will be zero. ✓
C) For a dataset with three classes, where two classes contain an equal number of instances and the third has none, the entropy will be maximum.
D) A dataset with five classes, where each class has a different number of instances, will always have a lower entropy than a dataset with five equally sized classes. ✓

a) $-\left(\frac{1}{4}\log\frac{1}{4} + \frac{1}{4}\log\frac{1}{4} + \frac{1}{4}\log\frac{1}{4} + \frac{1}{4}\log\frac{1}{4}\right) = 2$

b) $-\left[1\log 1 + 0\log 0 + 0\log 0 + \cdots\right] = 0$

c) max when equal dist.

d) max when equally distributed.

Consider a random forest model consisting of three small decision trees for a binary classification task. Each tree is built using a subset of features from the total available features in the dataset. Assume the dataset features are `X1`, `X2`, `X3`, and `X4`, and the outcome classes are `0` and `1`.

Here are brief descriptions of the decision trees:

- **Tree 1** uses features `X1` and `X2`, and has a depth of 3.
- **Tree 2** uses features `X3` and `X4`, and has a depth of 2.
- **Tree 3** uses features `X1`, `X2`, and `X3`, and has a depth of 4.

Based on the configuration of this random forest, identify which of the following statements are correct:

A) If one tree classifies a sample incorrectly, the other two trees' correct classifications can still ensure the correct final classification. ✓
B) The randomness in feature selection for each tree reduces the correlation between the trees, potentially increasing the forest's overall accuracy. ✓
C) Since Tree 3 uses more features and has greater depth, it will always dominate the classification decision in the random forest. ✗ all trees have equal importance
D) An increase in depth and number of features used in individual trees will always result in a proportional increase in the random forest's overall accuracy. ✗